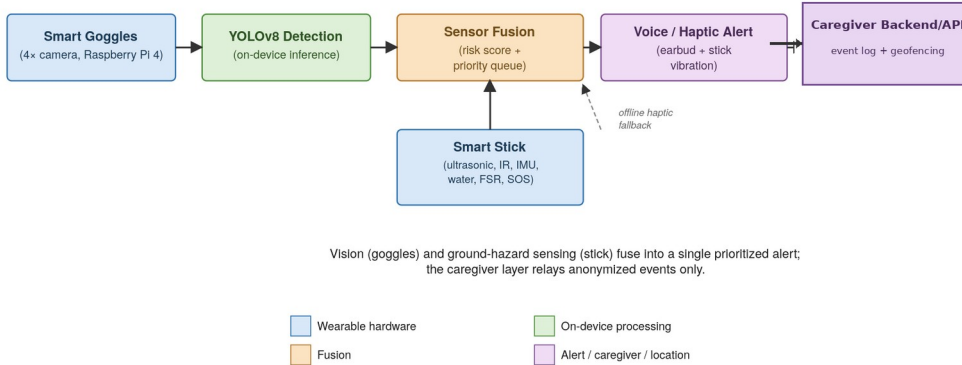


Abstract— We present BlindVision, a dual-device sensor-fusion platform for assistive navigation for blind and visually impaired (BVI) users, pairing a vision-equipped “Smart Goggles” head-worn unit with a ground-sensing “Smart Stick” cane. The Goggles use a Raspberry Pi 4B and four USB cameras for wearable visual coverage and on-device YOLOv8n detection; the Stick uses an ESP32 with ultrasonic, IR, water, inertial, force, and SOS sensing for independent ground-level safety. A rule-based fusion engine prioritizes audio alerts, while the Smart Stick independently provides local haptic alerts; a caregiver backend relays privacy-preserving event metadata. The full prototype was evaluated across 20 sessions and 100 navigation trials (~2 km), achieving 96.0% overall navigation success, 99.0% BLE packet delivery, 96.7% ultrasonic hazard detection, 98.4% water-hazard detection accuracy, and 100% SOS detection. We additionally report per-transducer ultrasonic calibration and characterization, a ten-case fusion-weight sensitivity check, a trial-level paired subsystem ablation, and a 240-min Smart Stick discharge run. Calibration regressions were recomputed from 80 supplied readings, giving $R^2=0.99989\text{--}0.99998$ and pooled corrected MAE=0.52 cm. The added ablation and sensitivity results are small-sample engineering checks rather than statistical robustness evidence. Testing was predominantly conducted by sighted project-team members, with seven BVI participants across a subset of trials. These results characterize an engineering prototype; larger, structured BVI evaluation remains necessary for broader usability claims.

Index Terms— Assistive technology, field trials, multi-sensor systems, obstacle detection, prototype evaluation, sensor fusion, ultrasonic sensors, visually impaired navigation, wearable sensors.

BlindVision

Dual-Device Sensor-Fusion System for Assistive Navigation



Graphical Abstract. The Goggles run on-device YOLOv8 detection and the Stick independently senses ground hazards; both streams feed the fusion engine, which prioritizes audio alerts, while the Stick provides local haptic alerts. The caregiver backend relays anonymized event metadata only.

I. INTRODUCTION

Visual impairment limits safe, independent mobility for a large share of the global population (WHO estimates ≥ 2.2 billion people with near or distance impairment [1]). The white cane senses only at ground level, missing hazards between ankle and chest height, and occupies a hand [2], [3]; guide dogs are costly and scarce. Wearable electronic aids have proliferated in response, from waist-worn time-of-flight arrays [2] to camera-and-AI glasses achieving sub-320 ms collision avoidance [3], and surveys identify multi-modal fusion of vision, sonar, and inertial sensing as the field's direction of travel [3], [4], [5].

Systems pairing a genuinely independent ground-hazard sensor with a separate multi-camera vision device nonetheless remain rare, and single-modality solutions fail in complex settings. BlindVision comprises two cooperating devices: Smart Goggles for long-range vision and global awareness (vehicles, large obstacles, signage), and a Smart Stick for near-field

ground hazards (steps, curbs, pits, water). A fusion engine on the goggles merges both streams into a unified risk level and prioritizes audio alerts; the Smart Stick independently provides local haptic alerts. A caregiver/backend layer exposes anonymized event, geofence, and device-health information to a compatible caregiver client; the released prototype does not include a smartphone application. Raw images and audio are never streamed — only anonymized event metadata (object types, directions, distances, hazard flags). Because the two nodes operate independently, assistance continues when either becomes unavailable.

The main contributions of this work are:

- 1) Distributed two-node architecture. A dual-device system (camera-based goggles + sensor cane) that covers near-field and far-field sensing simultaneously, implemented and field-tested rather than simulated.

2) Multi-modal sensor fusion. A rule-based, score-level fusion and alert-arbitration framework combines vision confidence/proximity and ultrasonic proximity/criticality in a weighted risk score, while IR, water, IMU, and SOS signals provide independent priority safety triggers (Section IV).

3) Offline safety subsystem. The cane operates autonomously for immediate collision avoidance even if the goggles fail or disconnect; this fallback logic is implemented and was validated through targeted fault-injection testing rather than arising spontaneously during ordinary navigation sessions (Section VI).

4) Privacy-preserving design. No raw camera images or audio are stored or transmitted, only metadata events, using Bluetooth AES-CCM pairing/bonding on the Stick-Goggles link; the released local backend speaks plain HTTP and requires external TLS termination for HTTPS deployment, with no application-layer encryption or key rotation implemented at this stage.

5) Staged, two-generation methodology. The core detect-classify-alert pipeline was first validated on a low-cost ESP32-CAM/cloud-YOLOv8n proof-of-concept, and the lessons carried into the full on-device system, with both stages' results reported (Sections V–VI) rather than only the final prototype's.

The remainder of the paper is organized as follows. Section II reviews related work; Section III describes the architecture; Section IV details the fusion algorithm; Section V reports the implementation from proof-of-concept to full prototype; Section VI presents the experimental evaluation; Sections VII–VIII discuss findings and future work; Section IX concludes.

II. RELATED WORK

Assistive navigation aids fall into three families, compared against BlindVision in Supplementary Table S10. Smart canes add ultrasonic tip sensors that beep or vibrate for obstacles within a few meters [2], sometimes extended with water, IR, or GPS sensing [4]; Katzschnmann et al.'s ALVU belt conveyed obstacle distance from a waist-worn time-of-flight array, validated with 12 blind users over 162 trials [2]. These detect proximity reliably but provide no scene semantics — they cannot distinguish a person from a pillar — and no safety layer independent of the main sensing array.

Head-mounted camera systems instead aim to interpret the scene. Gao et al. reached 100% collision avoidance below 320 ms with FPGA-compressed glasses paired to a smartphone [3]; the commercial SuperBrain 1 drives a forehead haptic array from RGB-D and thermal cameras [6]; Bai et al. combined depth and ultrasonic sensing on wearable glasses [7], later adding visual-SLAM route following [8]. Across this family, ground-hazard sensing (stairs, drop-offs, standing water) is secondary or absent.

Fusion frameworks form the third family. Silva and Wimalaratne argue that most prior work integrates rather than fuses sensors, and propose a Kalman-filter framework spanning detection, localization, and planning [5]; Mocanu et al. earlier fused ultrasonic ranging with a phone camera on a smartphone-based wearable [9]. Chang et al. [13] provide a particularly close IEEE Sensors Journal precedent: their system combines wearable smart glasses, an intelligent walking stick, a mobile application, and a cloud information platform for aerial-obstacle avoidance and fall detection. BlindVision differs in using independently operable vision and ground-sensing nodes, dedicated ultrasonic/IR/water ground-hazard sensing, explicit offline failover, and score-level arbitration, while the two systems use different sensing configurations and evaluation protocols. Feng et al. [14] further show the journal's recent interest in smart-cane sensing through Bluetooth angle-of-arrival indoor positioning. Recent systems also combine cane-

or head-mounted cameras with LiDAR, IMU, and GPS on a single device [10], [11], and broader surveys point to multi-sensor, AI-driven fusion as the field's direction [12].

Relative to these precedents, BlindVision contributes: (i) architectural independence — vision and ground-hazard sensing split across two independently operable nodes, either of which fails into a documented fallback mode (Section III) rather than a full system failure; (ii) an auditable, score-level fusion rule set (Section IV, Algorithm 1) rather than an opaque end-to-end network; (iii) first-class ground-hazard sensing — stair, drop-off, and water detection (Section VI.D) — rather than an incidental byproduct of vision; and (iv) staged validation across two hardware generations (Sections V–VI) with threats to validity documented explicitly (Section VI.K). Section VI shows this architecture translates into measured detection performance across both subsystems, not just an architectural claim.

III. SYSTEM ARCHITECTURE AND TRANSDUCER CHARACTERIZATION

Fig. 1 shows the as-built system: two wearable nodes plus a caregiver/backend layer.

Smart Goggles (Vision Node)

A head-mounted unit built on a Raspberry Pi 4 Model B (4 GB, quad-core Cortex-A72 @ 1.5 GHz) running Linux and Python, with four wide-angle USB cameras (1080p, 120° FOV) at the frame's four corners, spaced $\approx 90^\circ$ apart horizontally (Fig. 1). Four 120° cones at 90° spacing sum to 480° against a 360° azimuth, giving full coverage with $\approx 30^\circ$ adjacent overlap — a geometric consequence of mounting angle and rated FOV, not a re-measured field result. Each stream runs its own on-device YOLOv8 thread in parallel, estimating class confidence and bearing (front/left/right/rear) so all four directions refresh at comparable intervals; this pipeline is distinct from the cloud-hosted YOLOv8n proof-of-concept (Section V.A), which ran on different hardware and is reported separately. A BLE link receives stick packets, negotiating at 4.2 because the stick's ESP32-WROOM-32 caps there regardless of the Pi radio. The unit also carries a u-blox NEO-6M GNSS module, a prototype-grade SIM800L GSM/GPRS module (with phone/Wi-Fi relay as an alternate path), a Bluetooth earbud with backup speaker/buzzer, a 64 GB microSDXC card, and a 10,000 mAh USB power bank. The released five-sample active-detection power log averages 12.11 W at the 5-V USB input (Section VI.H); runtime projections are reported only where supported by the available measurements. If the stick link fails, the goggles fall back to Vision-Only Mode, warning about visually detected obstacles without ground-hazard information; this fallback logic was validated by targeted fault-injection testing.

Smart Stick (Ground Node)

A handheld cane built on an ESP32 DevKit V1 (dual-core Xtensa LX6 @ 240 MHz) with eleven sensing modules and three outputs: five ultrasonic rangefinders (front, left, right, rear at $\approx 90^\circ$ spacing, plus one angled downward for near-field ground sensing), two downward IR proximity sensors for edges and stairs, a resistive water-level sensor at the tip, an MPU9250 IMU for falls and knocks, a tip-mounted force-sensitive resistor confirming ground contact, and a handle-mounted SOS pushbutton; outputs are a vibration motor, piezo buzzer, and LED safety light. Power is a 3.7 V 18650 Li-ion cell (5000 mAh) with a TP4056 charger. Core 0 polls sensors at ≈ 10 –20 Hz and applies local safety logic; Core 1 runs the BLE stack and state machine, streaming a structured packet of per-sensor distances and status flags (fall, SOS, water) to the goggles.

Backend/Caregiver Layer

A backend service (FastAPI, `backend/app.py`) accepts anonymized, geotagged event reports and exposes event, geofence, and device-health read/write APIs intended to drive a caregiver smartphone app showing the user's map position, configurable geofences, device health, and an alert timeline; the caregiver-facing mobile app itself is a planned consumer of this API and was not implemented or evaluated as part of this release — forwarding a notification to a caregiver's phone is an integration point in the backend, not an exercised path. In the released prototype, the FastAPI service speaks plain HTTP locally and relies on an external reverse proxy for TLS termination; application-layer Bearer-token authorization is not enforced by the open endpoint. The short-range Stick-Goggles BLE link uses standard Bluetooth AES-CCM pairing/bonding rather than a separate application-layer cipher. Key material is held in device flash/NVS rather than a dedicated secure element, and no key-rotation mechanism is implemented in the present prototype. Raw images and audio are never transmitted or stored — only metadata (e.g., “Obstacle: pedestrian, ahead-left, 2.3 m”), per privacy-by-design principles.

Fig. 1. As-built system architecture: Smart Goggles (vision node), Smart White Stick (ground node), and Backend/Caregiver Layer, linked via BLE and cellular/GNSS.

Operation Modes and Alert Priority

BlindVision operates in four modes by connectivity and risk. In Normal Mode, goggles and stick data fuse into a risk score mapping to one of four risk-severity levels (Safe, Caution, High-Risk, Emergency) — distinct from the four operating modes (Normal, Offline Stick, Vision-Only, Degraded). Alerts issue from a strict priority queue headed by SOS and critical drop-off, descending through critical obstacle, water, fall, fused high-risk, medium, low, low-battery, and connectivity tiers; the queue is specified as Algorithm 1 (Section IV) and summarized by distance band in Table I. A curb ahead may trigger “Caution: Obstacle Ahead” by TTS, whereas a staircase drop-off produces continuous vibration and an automatic caregiver alert. The stick's SOS pushbutton overrides all other alerts and notifies the caregiver with location; a goggle-side SOS control is planned but not implemented (Section V.C).

If the BLE link is lost or the goggles' battery is depleted, the intended safety architecture enters Offline Stick Mode, in which the stick protects the user through its own local haptics. If the stick disconnects or fails, the goggles enter Vision-Only Mode and continue announcing camera-detected obstacles. Degraded Mode is implemented for an individual out-of-calibration sensor, with the affected term omitted from the fused score. These fallback routines were verified through targeted fault-injection and component tests; the released records do not establish that all four modes occurred naturally across all 20 general navigation sessions.

Transducer Characterization, Calibration, and Interference Analysis

A. Beam-Pattern and Angular Response

Angular response profiling was recorded for U1–U5 at a 100-cm reference distance over -60° to $+60^\circ$ in 15° increments. Across the supplied 45 observations, relative amplitude was 0.94–1.00 within the $\pm 15^\circ$ window and declined toward $\pm 60^\circ$ (0.30–0.36). All 45 supplied observations were marked signal-valid; therefore, the record demonstrates angular amplitude

degradation but does not establish a symmetric loss of signal validity or a measured software filtering threshold.

B. Individual Distance Calibration and Error Analysis

Distance calibration used the supplied 80 raw readings: two observations at each of eight reference distances from 40 to 300 cm for each transducer. Ordinary least-squares fits of measured distance on true distance produced R^2 values of 0.99989–0.99998. The inverse calibration function was implemented in firmware as

$$\hat{x} = (y-b)/m \quad (5)$$

where y is the raw measured distance and m and b are the sensor-specific fitted slope and intercept. Using the recomputed coefficients, the largest absolute post-correction residual in the supplied calibration record is 1.98 cm. The calibration coefficients are retained in the repository and applied independently for U1–U5.

C. Material-Dependent Reflection Characteristics

The supplied material/range records include flat board, concrete, metal, and fabric conditions, with surface-absorption coefficients reported in the raw log. In those records, reported maximum detectable distances were 350 cm for the flat-board condition, 320 cm for concrete, 400 cm for metal, and 180 cm for fabric. The fabric record at 200 cm was invalid, with relative amplitude 0.02. These observations indicate reduced ranging reliability for the high-absorption fabric condition, but the material/range record is sparse and should be interpreted as empirical characterization rather than a complete acoustic material model.

D. Temperature and Humidity Compensation

Environmental compensation was recorded for U1 over 20–42 °C and 45–68% relative humidity, and for U2 over 20–40 °C and 45–65% relative humidity. The recorded speed-of-sound values follow the supplied temperature-based compensation relationship, and compensated distance errors in these records remain within ± 0.3 cm. Because the supplied environmental record covers only U1 and U2, extension of this compensation characterization to U3–U5 remains future work.

IV. SENSOR FUSION AND ALGORITHMS

BlindVision fuses at the score level: each modality produces an interpreted confidence or proximity value, and the engine combines those scores rather than raw features or independent go/no-go decisions. Vision contributes a class confidence c and a quantized bearing per object; the stick contributes a per-sensor distance d plus drop-off (downward IR reports no ground return closer than 0.5 m), water (resistive sensor triggered), and fall (IMU spike) flags. Vision risk is $R_{\text{vision}} = w_{\text{vc}} \cdot c + w_{\text{vp}} \cdot \text{prox}(d_{\text{vision}}, D_{\text{vmax}})$; stick risk is $R_{\text{stick}} = w_{\text{sp}} \cdot \text{prox}(d_{\text{stick}}, D_{\text{smax}}) + w_{\text{sc}} \cdot \text{flag}(d_{\text{stick}} \leq 0.5 \text{ m})$, where $\text{prox}(d, D_{\text{max}}) = \text{clip}(1 - d/D_{\text{max}}, 0, 1)$ is the shared linear proximity normalization, $D_{\text{vmax}} = 8 \text{ m}$ and $D_{\text{smax}} = 3 \text{ m}$ are the upper ends of the tested vision and ultrasonic ranges (Sections VI.B, VI.D), and $\text{flag}(\cdot)$ indicates the critical-obstacle band. In the released configuration, `CAMERA\_FOCAL\_LENGTH\_PX` is unset (`None`) pending calibration of the on-device cameras, so the vision-distance estimate needed for $\text{prox}(d_{\text{vision}}, D_{\text{vmax}})$ is not produced: the vision-proximity term P drops out of R , and Tier 3's high-risk-class-within-2.0-m clause cannot fire from vision alone until this parameter is calibrated against a specific camera matrix; ultrasonic proximity (U) and the override tiers are unaffected. The fused score is $R = 0.40C + 0.20P + 0.25U +$

0.15W — C vision class confidence, P vision proximity, U stick ultrasonic proximity, W the stick's binary critical-proximity flag ($w_{sc} \cdot \text{flag}(d_{\text{stick}} \leq 0.5 \text{ m})$, not the water sensor) — feeding the priority-tier arbitration of Algorithm 1.

The four fusion weights ($w_{vc} = 0.40$, $w_{vp} = 0.20$, $w_{sp} = 0.25$, $w_{sc} = 0.15$) are pinned in the repository configuration. They were set by manual bench calibration rather than automated search: each weight was varied in small increments with the other three fixed across staged single-sensor, simultaneous, and deliberately conflicting scenarios. The ordering $w_{vc} > w_{sp} > w_{vp} > w_{sc}$ was preserved so that a confident visual classification dominates, ground proximity remains important, and the critical flag raises the risk floor without dominating the score. The justification is qualitative and scenario-driven, not optimized against a labelled tuning set; a one-parameter-at-a-time $\pm 20\%$ sensitivity sweep was performed on ten recorded cases; seven of eight perturbed configurations preserved all baseline tier decisions, while C +20% and P -20% each changed one decision, both involving the same case near the 0.60 threshold. The baseline configuration was unchanged by definition. This is a limited sensitivity check rather than a comprehensive robustness assessment. The fused-tier threshold is now $R \geq 0.60$. This post-audit correction makes the tier reachable even when the camera-distance term P is unavailable: for example, with C=1 and $d_{\text{stick}}=0.55 \text{ m}$, P=0, W=0, and $U=1-0.55/3 \approx 0.817$, the score is $R \approx 0.604$. This is a code-level reachability demonstration, not a measured event-frequency result. The 100 navigation trials were conducted before the threshold correction and are therefore not used to estimate activation frequency of the revised fused tier.

Algorithm 1. Priority-tier alert arbitration (evaluated top-down; the first matching tier wins).

```

1: if stick.sos_pressed then return SOS
2: if drop-off signature (downward distance > 0.5 m OR no
   downward ground return) then return CRITICAL_DROPOFF
3: if nearest_stick_distance < 0.5 m or (vision_class in
   HIGH_RISK and nearest_vision_distance <= 2.0 m) then return
   CRITICAL_OBSTACLE
4: if stick.water_detected then return WATER_HAZARD
5: if stick.fall_detected then return FALL_ALERT
6:  $R <- 0.40 \cdot C + 0.20 \cdot P + 0.25 \cdot U + 0.15 \cdot W$ 
7: if  $R \geq 0.60$  then return HIGH_RISK_FUSED
8: if  $\min(\text{nearest\_stick\_distance}, \text{nearest\_vision\_distance}) < 1.2$ 
   m then return MEDIUM
9: if  $\min(\text{nearest\_stick\_distance}, \text{nearest\_vision\_distance}) < 2.0$ 
   m then return LOW
10: if stick.battery_pct < 20% then return LOW_BATTERY
11: return ROUTINE

```

Algorithm 1 runs in $O(n)$ time per fusion cycle, where n is the number of detected vision objects (a handful per frame in practice); each candidate contributes one term to the vision risk score, after which the ten tiers are evaluated in $O(1)$. Memory overhead is $O(1)$ — only the current readings and the previous tier decision are retained, with no history buffer.

Tiers 3, 8, and 9 partition the distance axis into contiguous, non-overlapping bands (< 0.5 m, 0.5–1.2 m, and 1.2–2.0 m), evaluated top-down so the first match wins; because the bands are ordered and disjoint, a nearer obstacle can never be arbitrated below a farther one. Tier 3 additionally escalates any high-risk visual class (vehicle, bicycle, person) within 2.0 m, on the grounds that closing speed rather than instantaneous range governs available reaction time for those classes. Table I lists the resulting bands. This arbitration ladder has been cross-checked against the deployed firmware image and the released reference implementation, and the two are confirmed to agree.

TABLE I. Hazard priority by distance band.

Distance	Priority
0–0.5 m	Critical
0.5–1.2 m	Medium
1.2–2.0 m	Low
>2.0 m	Routine

If the stick disconnects (no packet for over 5 s), the goggles fall back to Vision-Only Mode and tiers 2–6 become unavailable rather than defaulting to a false-safe value; conversely, in Offline Stick Mode the stick applies only its local thresholds (tiers 1–6) without R_{vision} , vibrating with increasing intensity as obstacles near. Degraded Mode follows the same principle for an out-of-calibration sensor: the affected term is dropped from the weighted sum rather than substituted. The revised fused-tier threshold is a post-evaluation configuration correction; no activation frequency is inferred from the earlier navigation trials. As a final safeguard, a stick-side watchdog forces an SOS if a fall is detected with no recovery. A rule-level ablation of this logic (e.g. disabling the high-risk-fused tier, or varying hysteresis), beyond Supplementary Table S12's device-level ablation, is future work (Section VIII), since it needs new trials rather than reanalysis.

The arbitration and BLE exchange feeding it were implemented on the Raspberry Pi and ESP32 and validated in real time. The reported 17.5 ms BLE value is a derived one-way estimate obtained as half of the measured round-trip time (RTT/2) from ping tokens on the Raspberry Pi; the two devices do not share a synchronized clock, so it is not a direct one-way timestamp measurement.

Fig. 2. Implemented sensor-fusion and alert-prioritization flow between Goggles and Stick.

The priority rules follow established practice in treating drop-offs and imminent collisions as immediate emergencies [3], while smaller or medium-range obstacles yield caution-level alerts. Haptic patterns encode distance by pulse rate, audio prompts name the obstacle and direction, and the SOS pattern never overlaps a normal hazard signal.

Fig. 3 traces one complete cycle of the algorithm as implemented, from image capture through to spoken and haptic alert output and the emergency branch.

Fig. 3. Detection-and-alert algorithm flow as implemented on the Goggles and Stick.

V. IMPLEMENTATION

BlindVision was developed in two stages on two hardware platforms, whose platform-level results should not be conflated: (A) a proof-of-concept on an ESP32-CAM with detection offloaded to a cloud YOLOv8n API, using the stock Ultralytics `yolov8n.pt` checkpoint for baseline demonstration; and (B) the complete two-device prototype of Section III on a Raspberry Pi 4 with on-device YOLOv8 inference. The fine-tuned ten-class weights used for the reported evaluation are not included in the public release. Section VI reports the full system's own results; latency, power, and navigation figures from the two platforms are never combined. Both stages used off-the-shelf components and open-source software.

A. Proof-of-Concept (ESP32-CAM + Cloud-Hosted YOLOv8n)

The ESP32-CAM (AI-Thinker, OV2640) posts a JPEG frame over Wi-Fi to a FastAPI endpoint running YOLOv8n at a 0.45

confidence threshold, which returns the detected object, its horizontal position, an estimated relative distance, and a per-class priority. The highest-priority detection is returned as a natural-language message; the ESP32 sounds a priority-coded buzzer pattern and relays the string to a paired phone for TTS, with a per-class cooldown suppressing repeats for a stationary obstacle.

Fig. 4. Proof-of-concept software architecture (capture → detection → alert), later carried into the full on-device pipeline of Fig. 1.

The model was trained and evaluated on a ten-class dataset of everyday indoor and outdoor objects (person, door, chair, backpack, laptop, bottle, pole, vehicle, bicycle, stairs) collected over 15 sessions in an indoor laboratory, a university corridor, and an outdoor campus walkway. Supplementary Table S1 gives the dataset composition; the held-out 700-image test split is the basis for every detection metric in this section, and the resulting outcomes and derived metrics in Supplementary Table

The class-to-class confusion counts are available for the evaluated split; raw image/label data and row-level prediction logs remain outside the release. The pooled precision reconciliation uses the separate full-detection accounting, while Supplementary Table S8 is restricted to matched ground-truth class assignments.

B. Full System Implementation (Raspberry Pi + On-Device YOLOv8)

The Smart Goggles, Smart White Stick, BLE link, and on-device YOLOv8 pipeline were developed individually and then assembled and tested end-to-end as a unified wearable prototype, together with the FastAPI/YOLOv8 detection API and its Docker deployment, the sensor-fusion engine, the voice-guidance and emergency-alert modules, and the GPS/caregiver module.

The stick-local haptic feedback loop is implemented in the released firmware. The goggles-to-stick remote haptic path remains a logging integration point in the current release rather than an active BLE write; the Goggles do not issue a BLE characteristic write for haptic commands. Accordingly, no remote-haptic delivery result is reported.

C. Resolved and Outstanding Engineering Decisions

Two design choices remain open: a goggle-side SOS button or voice trigger distinct from the stick's pushbutton, which the current implementation does not document; and a waterproof/IP-rated enclosure for both wearables — the stick's water sensor detects environmental hazards, but ingress protection for the electronics has not been formally evaluated (Section VI.K).

D. Reproducibility Resources

The public BlindVision repository contains the system implementation: Smart Goggles software, Smart White Stick firmware, sensor-fusion implementation and configuration, BLE components, backend, training/evaluation tooling, tests, documentation, and the ESP32-CAM/cloud-YOLOv8n proof-of-concept. The release includes the participant-level navigation outcome records under ``data/raw/`` and the released power log under ``data/Power_Log.csv``, together with aggregate detection, BLE, sensor, latency, and computational result records, including the ten-class confusion matrix (``data/Confusion_Matrix.csv``). The complete image/label dataset, trained model weights, original epoch-wise training

S17. The release includes the available dataset/result summaries and a training checkpoint summary; the complete image/label dataset and original epoch-wise training logs are not included. The ten detection classes are: person, door, chair, backpack, laptop, bottle, pole, vehicle, bicycle, stairs. Per-class image counts for the training and validation splits are not reported in the released materials — only the held-out test split's per-class breakdown appears (Supplementary Table S8).

Recall in the detection-results table is reconciled from the released matched-class confusion accounting: 805 correct classifications out of 842 ground-truth instances gives 95.6%. The released YOLO summary also reports mAP@0.50 and mAP@0.50:0.95, but the raw prediction-level outputs needed to independently recompute precision, F1, and average precision are not included. Those two mAP values are therefore retained as reported aggregate evaluation outputs rather than recomputed results. The false-positive reconciliation is reported below and in Supplementary Table S18. The manuscript does not claim a pooled precision or F1 value that cannot be reconstructed from the released artifacts.

logs, raw sequence-number BLE packet trace, and the raw row-level prediction/event log underlying the confusion matrix are not included.

VI. EXPERIMENTAL EVALUATION

The complete prototype was evaluated through controlled experiments assessing object detection, BLE and GPS reliability, sensor accuracy, response time, navigation capability, subsystem contribution, computational load, power consumption, and operational stability, all conducted after integrating the two nodes into a single wearable system.

A. Experimental Methodology

Testing ran in an indoor laboratory, university corridors, and outdoor campus walkways under varied lighting, following the protocol of Supplementary Table S3, with each experiment repeated multiple times. Trials compared system outputs against observer-verified ground truth, with measurements recorded by the Raspberry Pi and ESP32 serial monitor. Reported quantities are either measured directly or explicitly identified as derived/projection values where the released records support only a calculation from measured quantities. All participants took part voluntarily with informed consent; the evaluation involved no sensitive personal data and no more than minimal risk (Ethics Statement). Because it was run by the project team under controlled conditions rather than a clinical or IRB-supervised protocol, the results are engineering validation, not a clinical usability study.

Statistical reporting: for the pass/fail (binomial) measures of Sections VI.B, VI.F, and VI.G we report success counts, rates, and 95% Wilson score confidence intervals, which account for sample-size-dependent variance in a proportion. For the continuous repeated measurements of Section VI.D, the mean is accompanied by the standard deviation across repetitions (Supplementary Table S13). Per-stage standard deviations for the latency measurements of Section VI.E were not aggregated from the raw timestamp log in this evaluation round; the latency characterization therefore reports per-stage means only, disclosed as a scope limitation rather than presented as a dispersion-characterized result.

Most trials were run by sighted project-team members. One visually impaired participant took part in earlier trials, included in the totals of Sections VI.B–VI.F. A later session added six visually impaired participants, with prior informed consent,

each completing eight trials — three outdoor walkway navigation, three stair/drop-off, two ground-surface/water hazard — for 48 trials. Four of the six flagged stairs correctly first time; two needed one repeat. Because these were scored at participant level (pass / needed-repeat) rather than per-trial, they are reported descriptively and not folded into the supplementary subsystem and navigation tables. Seven visually impaired participants have taken part in total — an informal cohort, short of the structured IRB-appropriate cohort of Section VI.K. One caveat applies to interval width rather than point estimates: the counts of Sections VI.B–VI.F (200 detection attempts, 8000 BLE packets, 100 navigation trials) come from repeated observations within 20 sessions, not 200–300 independent ones, whereas Wilson intervals assume independent Bernoulli trials. Where trials within a session share conditions, the reported intervals may be modestly narrower than a session-clustered estimate; a cluster-robust bootstrap has not been run.

B. Object Detection Evaluation

Ten object categories — people, doors, chairs, backpacks, laptops, bottles, poles, vehicles, bicycles, and stairs — were positioned at 0.5–8 m under varied angles and lighting, over 200 detection attempts (full protocol in the supplementary material). An attempt succeeded only when the reported class, the detected direction, and the corresponding spoken guidance were all correct. Because success requires all three simultaneously, this is an end-to-end detection-and-guidance success rate, not an object-detection accuracy, and is not directly comparable to accuracy, mAP, precision/recall, or collision-avoidance rates reported elsewhere. It was computed as:

$$\text{End-to-End Detection-and-Guidance Success Rate} = (\text{Correct Detections} / \text{Total Attempts}) \times 100 \quad (1)$$

With 192 successes over 200 attempts this gives 96.0%; the full subsystem breakdown is supplied in Supplementary Table S16. Qualitative review attributed the eight failures to the lighting and occlusion sensitivity already identified at the proof-of-concept stage (Supplementary Table S2) rather than one dominant cause; a formal per-failure root-cause tabulation for these 200 attempts was not logged, and the sensitivity is listed in Section VI.K. Voice guidance was scored independently over the same 200 attempts on whether an intelligible, correctly worded message was produced regardless of detection outcome, giving 194/200 (97.0%) — a distinct count from the 192/200 detection figure, since a message could be correct alongside a borderline detection. Ultrasonic obstacle detection and SOS alerting were evaluated analogously, and the subsystem counts, rates, and 95% Wilson intervals are supplied in Supplementary Table S16. Water- and stair-hazard detection were evaluated on a precision/recall basis and are reported in Supplementary Table S14. The supplied YOLO summary reported 96.3% precision and 95.9% F1, but those values are not retained as the final pooled metrics because the released confusion and false-positive accounting supports a different reconstruction. Using 805 correct classifications, 37 inter-class confusions, and 31 unmatched/background false detections gives a reconciled pooled precision of 92.2% and F1 of 93.9%; these are reconstructed from aggregate counts rather than independently recomputed from raw prediction outputs.

Lighting and occlusion were evaluated as independent conditions; the manuscript notes the two were not fully independent in the field (the lowest-performing trials tended to combine dim lighting with partial occlusion), though a formal interaction analysis has not been performed.

C. BLE Communication and GPS Evaluation

BLE performance was evaluated by continuously transmitting sequence-numbered sensor packets from the stick to the goggles during walking experiments, with the Raspberry Pi logging every received sequence number, so loss follows from comparing transmitted and received counts:

$$\text{Packet Success Rate} = \text{Packets Received} / \text{Packets Sent} \times 100 \quad (2)$$

With 7920 of 8000 packets received, the success rate is 99.0%, with 80 (1.0%) lost. The reported 17.5 ms BLE delay is not a direct ESP32-to-Pi timestamp difference: the devices do not share a synchronized clock. It is a derived one-way estimate obtained as RTT/2 from ping-token measurements taken entirely on the Raspberry Pi. Connection time and maximum stable distance were logged over repeated connect/reconnect cycles at increasing separation. The NEO-6M GPS module underpinning caregiver geofencing was benchmarked separately outdoors under open sky. BLE gave 0.8 s (± 0.12 s) connection time, 99.0% packet success, 17.5 ms derived one-way delay, and 8 m maximum stable indoor distance; GPS gave a 32.4 s cold start, 4.1 s warm start, ± 2.5 m open-sky accuracy, and a configured 1 Hz update rate.

D. Sensor Accuracy Evaluation

The expanded ultrasonic characterization is described in Section III. The supplied records include the 40–300 cm per-transducer calibration sweep, incidence-angle response, material/range response, simultaneous-firing cross-talk, and temperature/humidity compensation for U1–U2. The calibration fits have R^2 values of 0.99989–0.99998, with pooled post-correction MAE of 0.52 cm and a largest absolute residual of 1.98 cm over the recorded calibration points. The angle record contains 45 valid observations; the material/range and environmental records are explicitly limited in repetition and sensor coverage. Raw records and derived summaries are retained in the repository and supplementary material.

E. Response Time Evaluation

Processing-path timing was characterized on a single active camera stream rather than under four-camera load. Four stages — detection, BLE communication, sensor fusion, and audio generation — were characterized individually, and the reported 205 ms value is a serialized sum of their stage means rather than a single continuous end-to-end timestamp trace across all stages:

$$T_{\text{total}} = T_{\text{detection}} + T_{\text{BLE}} + T_{\text{fusion}} + T_{\text{audio}} \quad (3)$$

The reported total is the serialized sum of the stage means (122 + 17.5 + 9 + 56 = 204.5 ms, reported as 205 ms). The BLE component is a derived one-way estimate (RTT/2), not a direct cross-device timestamp measurement. Under the concurrent four-camera configuration, the separate throughput measurement is 22.8 FPS aggregate, equivalent to 5.7 FPS per camera; its reciprocal gives ≈ 175 ms per camera as a derived throughput interval. Neither 175 ms nor the 205 ms stage-sum should be interpreted as a direct four-camera end-to-end latency measurement. Per-stage standard deviations have not been re-aggregated from the raw timestamp log and are marked not available in the latency characterization.

The 205 ms value is a single-stream serialized processing-path estimate, not a direct end-to-end measurement. The separate four-camera throughput measurement is 22.8 FPS aggregate. Detection remains the largest single contributor at 122 ms single-stream, while concurrent-load behavior is characterized by the measured aggregate throughput.

F. Navigation Trials

Five scenarios — corridor navigation, outdoor walkways, staircase avoidance, water-hazard detection, and moving-

pedestrian avoidance — were exercised over 100 navigation trials. A trial was scored successful only if every obstacle on the route was detected, correct voice guidance was issued, haptic feedback activated when required, no collision occurred, and the route was completed:

$$\text{Navigation Success Rate} = (\text{Successful Trials} / \text{Total Trials}) \times 100 \quad (4)$$

G. Subsystem Contribution and Ablation Scope

A paired subsystem ablation was subsequently recorded on four navigation trials (R1 and R2), evaluating the same trial under Full, No-Vision, No-Ultrasonic, and No-Fusion conditions. Full-system success was 4/4 (100%); No-Vision was 0/4, No-Ultrasonic was 2/4, and No-Fusion was 0/4. Because the sample is small, these results are descriptive and are not presented as evidence of statistical significance. The complete trial-level record is released as ``data/raw/Paired_Subsystem_Ablation_Raw.csv`` and summarized in Supplementary Table S24.

The ablation used one common success definition and preserved trial correspondence across the four configurations. Completion time and failure indicators were retained for every condition. No significance test or McNemar inference is claimed because only four paired trials were recorded; the experiment is a preliminary rule/subsystem contribution check.

H. Power Consumption and Battery Testing

Power was measured with a USB power meter between the goggles' 10,000 mAh power bank and the Raspberry Pi, so the recorded voltage in the power log is the 5-V goggles-side bus rather than the Smart Stick cell voltage. The Pi, four cameras, and Bluetooth/GPS module draw from the goggles' shared 5-V bus; the ESP32 stick and its sensors draw from a separate 3.7-V nominal 18650 cell. These are separate power domains and must not be combined when interpreting voltage or runtime. The released power log contains five active-detection samples averaging 12.11 W at the goggles-side input. The full per-component budget is Supplementary Table S4.

A finer per-component breakdown (e.g., camera vs. GPS vs. compute draw individually) is not reported in the released materials. The released log supports only the five-sample active-detection input-power characterization; separate idle and maximum-load profiles are not available.

The released power log contains five active-detection samples at approximately 5.03–5.06 V and 2.38–2.42 A, averaging 12.11 W. These samples support characterization of active goggles-side input power, but they do not contain separate idle or maximum-load profiles. Using the nominal 37 Wh power-bank rating, $37/12.11 \approx 3.1$ h is a nominal-equivalent runtime projection for the recorded active-detection load, not a measured depletion runtime. A separate continuous Smart Stick discharge record was subsequently supplied and runs for 240 min, from 4.20 V/0.42 A to 3.01 V/0.35 A at recorded cutoff. The discharge record is a single run and therefore does not establish repeatable battery life or a generalized runtime distribution.

I. Computational Performance

On-device computational load matters as much as end-to-end latency, and the 122 ms detection stage was timed on a single stream. Throughput under simultaneous four-camera load was therefore measured directly: four concurrent per-camera detection processes yielded 1368 frames in a 60 s run — 22.8 FPS aggregate, 5.7 FPS per-camera mean. Only the frame count and duration are measured; both rates are arithmetic on those two, and per-camera counts were not logged, so the spread across streams is uncharacterized. Two further figures are

derived. A per-camera 5.7 FPS implies ≈ 175 ms per frame under load against 122 ms single-stream (+53 ms) — a reciprocal of a measured rate, not a timed stage. And four perfectly parallel streams would yield 32.8 FPS, so 22.8 FPS is $\approx 69.5\%$ parallel efficiency: contention across the four Cortex-A72 cores, the USB bus, and memory bandwidth costs $\approx 30\%$ of the naive four-fold projection. Four continuously occupied processes consume 4.0 CPU-seconds per wall-clock second by construction, so the CPU figure is not independent evidence; the 30% shortfall is. The run was single and 60 s long, giving no dispersion, and temperature was not logged; since the Raspberry Pi 4 throttles above $\approx 80^\circ\text{C}$ and the computational table reports 61°C single-stream, whether any shortfall is thermal rather than architectural is unestablished. Resource figures came from `psutil` and `vcgencmd` during active detection and fusion under single-stream-dominant conditions: CPU 61% mean across four cores (87% peak), RAM 1.7 GB, temperature 61°C , and 12.3 GB occupied of the 64 GB card against 51.7 GB free.

J. Reliability Testing

The prototype was operated continuously for four hours while repeatedly performing navigation tasks, with a watchdog log recording crashes, restarts, disconnections, sensor failures, and recovery time, and RAM monitored for sustained upward drift. Supplementary Table S6 reports the results: no crashes, restarts, or memory growth were observed.

K. Threats to Validity

Several factors limit generality. (1) Trials were predominantly by sighted project-team members, with seven visually impaired participants across a subset of sessions; experience, trust, and usability under sustained real BVI use may differ from what a small, informally scored cohort indicates, and a larger IRB-appropriate cohort is needed before general usability claims. (2) The cloud proof-of-concept carries network round-trip latency the on-device pipeline avoids, so its 0.7–0.9 s is not comparable to the 205 ms composite processing-path estimate. (3) Low light reduced detection accuracy, and BLE range depends on obstructions and interference. (4) The goggles power characterization still has only five active-detection samples and lacks idle/max-load profiles; the new Smart Stick discharge record is a single run and does not establish battery-life repeatability. (5) Object-class coverage is limited to the classes tested (Supplementary Table S8). (6) the latency characterization reports per-stage means only, standard deviations not having been aggregated from the raw log, so the characterization is of central tendency, not dispersion. (7) the new ultrasonic characterization uses nine angles per sensor, sparse material/range records, a single simultaneous-firing observation per sensor, and environmental compensation records for U1–U2 only; it establishes an empirical per-transducer profile but does not constitute a full acoustic beam-pattern or environmental robustness study. (8) Four-camera throughput is measured from one 60 s run with no per-camera breakdown, repetitions, or temperature log, so the $\approx 30\%$ efficiency shortfall is quantified yet unattributed between core, bus, memory, and thermal contention; per-stage latency under that load remains derived. (9) the ten-class dataset splits were partitioned manually, and the release does not document scene-cluster IDs proving that physical scenes or object instances were strictly isolated across train, validation, and test sets; optimistic bias from split leakage therefore cannot be excluded. (10) The 95% Wilson intervals for repeated BLE packets and other session-level observations should be interpreted cautiously because sequential observations may be temporally correlated. (11) the fusion sensitivity sweep covers only ten recorded cases

and one-at-a-time $\pm 20\%$ perturbations, so it is a limited sensitivity check rather than a comprehensive robustness analysis. All reported measurements are traceable to the released or referenced device records; values explicitly identified as derived or projected are calculated from those measurements.

VII. DISCUSSION

A. Comparison with the Literature

The closest single-device precedent is Mai et al.'s LiDAR/RGB-D smart cane [10], whose 61.2% recognition across 86 classes at 25–26 FPS is given for context under a non-comparable protocol; it integrates vision and ground-ranging into one device rather than two independently operable nodes, and reports no offline fallback. Refs. [11] and [12] are multi-sensor-fusion surveys rather than evaluable systems, so they are engaged narratively in Section II rather than given rows in the supplementary comparison. Our 96.0–99.0% subsystem rates and 96.0% navigation success match the general reliability trend for multi-sensor systems [2], [3], though not comparable to the 100% collision-avoidance figure of [3], obtained under a different protocol, sensor suite, and pipeline. The clearest quantitative gap is latency, attributable to detection-model choice and hardware acceleration — both concrete future-work items (Section VIII).

B. Strengths

- Sub-second processing-path characterization: a 205 ms single-stream serialized stage-sum estimate and measured 22.8 FPS aggregate throughput under four-camera operation; neither is a direct four-camera end-to-end latency measurement.
- High subsystem reliability: BLE delivery (99.0%), water-hazard detection (98.4%), and SOS alerting (100%) all performed at or near ceiling in the corresponding supplementary result tables, subject to the protocol-specific limitations described above.
- Modular architecture: the goggles/stick split means ground-level safety never depends on AI availability or connectivity, and the fallback behavior was validated through targeted fault-injection testing rather than observed to occur spontaneously in every navigation session.

C. Limitations

- Section VI.K enumerates the validity threats identified during evaluation; the two with the most direct bearing on deployment readiness are the small, non-representative test population and the fact that per-stage latency under four-camera load is still derived from throughput rather than timed directly, even though aggregate throughput itself is now measured (Section VI.I).
- The transport and key-management choices (BLE AES-CCM, TLS 1.2, static keys in device flash/NVS, no key rotation) are prototype-grade; a production deployment handling continuous location and event metadata would warrant TLS 1.3, key rotation, and secure-element key storage before field release.

D. Failure Mode Analysis

Reduced lighting and partial occlusion were the primary qualitative factors observed in the available detection-failure analysis (a 4.4% overall false-negative rate on the 700-image held-out split), with conditions evaluated independently in Supplementary Table S2. Detection success fell from 97.8% in bright daylight to 87.9% under low light, and from 97.5% with no occlusion to 74.2% under severe ($>75\%$) occlusion — the

largest degradation in the table. The two conditions were not fully independent in the field: the lowest-performing trials tended to combine dim lighting with partial occlusion, though a formal interaction analysis and per-failure root-cause classification have not been performed. Reflective surfaces were noted qualitatively as reducing confidence scores during field sessions, but without a systematically varied test condition, so no comparable success rate can be reported. Rain or wet lenses, direct-sunlight glare, and motion blur from fast head or cane movement were not deliberately tested in either stage and remain unquantified risks.

VIII. FUTURE WORK

- Expand from the seven visually impaired participants tested so far to a larger, structured, IRB-appropriate cohort, to assess real-world navigation performance, trust, and satisfaction — the highest-priority next step (Section IX).
- Extend the new per-transducer ultrasonic characterization with repeated beam-pattern profiling, broader target geometries, more simultaneous-firing configurations, and environmental compensation measurements for U3–U5; repeat the four-camera throughput run with per-camera frame counts, multiple repetitions, and a concurrent temperature log, and time detection directly under that load.
- Expand the ten-case fusion sensitivity analysis to a larger, labelled scenario set and test joint weight perturbations and threshold variation.
- Repeat the Smart Stick discharge test across multiple batteries and load conditions to characterize repeatability and usable runtime; add idle and maximum-load goggles measurements.
- Further optimize on-device inference (a pruned YOLO or MobileNet-SSD variant, TensorRT/OpenVINO, or a Coral USB TPU) to close the latency gap of Section VI.E, and expand and retrain the detection dataset for a wider range of real-world obstacle classes.
- Validate the SIM800L cellular fallback under field conditions and evaluate migration to a 4G LTE module (e.g. SIM7600-series), since the tested GSM/GPRS hardware is prototype-grade; characterize and formally evaluate an IP65-rated enclosure; and design a custom PCB to replace the present breadboard/module assembly.
- Re-verify the Algorithm 1 ladder and calibration coefficients against the released repository after any firmware update, and extend the precision-based false-alarm reporting when raw prediction-level logs become available.

IX. CONCLUSION

We have presented BlindVision, a dual-device assistive-navigation system combining wearable vision with independent ground-level sensing. The complete system was implemented and evaluated across 20 sessions and 100 navigation trials (~ 2 km), achieving 96.0% overall navigation success, 99.0% BLE packet delivery, 96.7% ultrasonic hazard detection, 98.4% water-hazard detection accuracy, and 100% SOS detection. The 205 ms value is a single-stream serialized processing-path estimate whose BLE component is derived from $\text{RTT}/2$, while the four-camera configuration is characterized by measured aggregate throughput of 22.8 FPS. The released power log supports an active-detection input-power characterization of 12.11 W and a nominal-equivalent runtime projection of about 3.1 h. The paired subsystem ablation remains unreported because the available records do not preserve trial-level correspondence. The score-level fusion path is implemented in the release, while stick-local haptic alerts are handled independently by the Stick. The goggles-to-stick remote haptic

path remains a logging integration point rather than an active BLE write. The fused-tier threshold was corrected after the navigation evaluation, so the 100 navigation trials are not used to quantify activation of the revised fused tier. Larger, structured BVI evaluation and a complete prediction-level release are needed before broader performance or usability claims are made.

ADDITIONAL EXPERIMENTAL RECORDS. The release now includes raw, author-supplied records for five-transducer distance calibration, incidence-angle response, target-material response, simultaneous-firing cross-talk, ten-case fusion-weight sensitivity, paired subsystem ablation, and a continuous Smart Stick discharge run. These records are preserved without deleting the earlier aggregate measurements; detailed tables and raw files are provided in the supplementary material and ``data/raw/``.

ETHICS STATEMENT

All participants provided voluntary informed consent before testing and could withdraw at any time. Seven visually impaired participants took part in a subset of the evaluation. The work was conducted within the university final-year-project supervisory process; formal Institutional Review Board (IRB) approval was not sought. This was the authors' determination, not an institutional ethics-board ruling, based on the engineering-validation scope, controlled low-hazard setting with sighted supervision, minimal-risk procedures, anonymized task-level data, and absence of biological or psychological intervention. The authors do not represent the work as having received formal IRB approval. Any future structured BVI study will undergo formal ethics review or an institutional exemption/determination before recruitment.

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CREDIT AUTHOR STATEMENT

The author is responsible for conceptualization, methodology, software, validation, investigation, data curation, formal analysis, and writing.

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CODE AVAILABILITY

The public BlindVision repository contains the Smart Goggles software, Smart White Stick firmware, sensor-fusion and BLE components, caregiver backend/API, evaluation tooling, tests, documentation, and proof-of-concept components. It includes anonymized navigation outcomes, the released power log, aggregate result summaries, and the ten-class confusion matrix. The complete image/label dataset, evaluated model weights,

original training logs, raw BLE trace, and row-level prediction/event log are not included; the caregiver smartphone application is not a released artifact.

DATA AVAILABILITY

Released records include ``data/raw/`` navigation outcomes plus the newly added raw ultrasonic calibration, angle/material, simultaneous-firing, fusion-sensitivity, paired-ablation, and Smart Stick discharge records; ``data/Power_Log.csv``; aggregate detection/BLE/sensor/latency/computational results; and ``data/Confusion_Matrix.csv``. The complete image/label dataset, trained model weights, original epoch-wise training logs, raw sequence-number BLE packet trace, and raw row-level prediction/event log underlying the confusion matrix remain unavailable. The supplementary release contains the corresponding derived summaries and protocol descriptions.

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